

Corrected calculations for network paper

(Dated: June 9, 2026)

PRELIMINARIES

These are preliminary identities that can be derived by using the finite series identities in PRX. I have derived the following for $|m| < N$ and $m \neq 0$

$$\frac{1}{N} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} = 1 \quad (1)$$

Another identity exploits the fact that

$$\begin{aligned} [1 - (-1)^k] \sin \left[\frac{|m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right] &= \frac{1}{2} \left\{ \cos \left[\frac{(|m| - 1)\pi k}{N} \right] - \cos \left[\frac{(|m| + 1)\pi k}{N} \right] \right. \\ &\quad \left. - \frac{1}{2} \left[\cos \left[\frac{(N - |m| + 1)\pi k}{N} \right] + \cos \left[\frac{(N + |m| - 1)\pi k}{N} \right] - \cos \left[\frac{(N - |m| - 1)\pi k}{N} \right] - \cos \left[\frac{(N + |m| + 1)\pi k}{N} \right] \right] \right\} \end{aligned}$$

for a generic σ we have

$$\begin{aligned} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{\sigma - \cos \left[\frac{\pi k}{N} \right]} &= \frac{1}{2} \sum_{k=1}^N \frac{\cos \left[\frac{(|m| - 1)\pi k}{N} \right] - \cos \left[\frac{(|m| + 1)\pi k}{N} \right]}{\sigma - \cos \left[\frac{\pi k}{N} \right]} \\ &\quad - \frac{1}{4} \sum_{k=1}^{N-1} \frac{\cos \left[\frac{(N - |m| + 1)\pi k}{N} \right] + \cos \left[\frac{(N + |m| - 1)\pi k}{N} \right] - \cos \left[\frac{(N - |m| - 1)\pi k}{N} \right] - \cos \left[\frac{(N + |m| + 1)\pi k}{N} \right]}{\sigma - \cos \left[\frac{\pi k}{N} \right]} \\ &= \frac{N}{2(\sigma^2 - 1)U_{N-1}(\sigma)} \{T_{N-|m|+1}(\sigma) - T_{N-|m|-1}(\sigma) - T_{|m|-1}(\sigma) + T_{|m|+1}(\sigma)\} \end{aligned} \quad (2)$$

If $\sigma = \cos \left[\frac{2\pi r}{N} \right]$ one needs to use de l'Hospital rule to evaluate the expression above, which gives

$$\begin{aligned} \lim_{\sigma \rightarrow \cos \left[\frac{2\pi r}{N} \right]} \frac{N}{2(\sigma^2 - 1)U_{N-1}(\sigma)} \{T_{N-|m|+1}(\sigma) - T_{N-|m|-1}(\sigma) - T_{|m|-1}(\sigma) + T_{|m|+1}(\sigma)\} \\ = \cos \left[\frac{2\pi r|m|}{N} \right], \end{aligned} \quad (3)$$

in other words we have the general identity for $0 < |m| < N$ and $r = 1, \dots, N-1$,

$$\frac{1}{N} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{\cos \left[\frac{2\pi r}{N} \right] - \cos \left[\frac{\pi k}{N} \right]} = \cos \left[\frac{2\pi r|m|}{N} \right], \quad (4)$$

The third relevant identity for $|m| \leq N$ and integer k is given by

$$\frac{1}{N} \sum_{r=1}^{N-1} \frac{\cos \left[\frac{2\pi r m}{N} \right]}{\cos \left[\frac{(2k-1)\pi}{N} \right] - \cos \left[\frac{2\pi r}{N} \right]} = \frac{1}{N \left\{ 1 - \cos \left[\frac{(2k-1)\pi}{N} \right] \right\}} - \frac{\sin \left[\frac{(2k-1)\pi|m|}{N} \right]}{\sin \left[\frac{(2k-1)\pi}{N} \right]} \quad (5)$$

alternatively

$$\frac{1}{N} \sum_{r=1}^{N-1} [1 - (-1)^k] \frac{\cos \left[\frac{2\pi r m}{N} \right]}{\cos \left[\frac{k\pi}{N} \right] - \cos \left[\frac{2\pi r}{N} \right]} = [1 - (-1)^k] \left\{ \frac{1}{N \left\{ 1 - \cos \left[\frac{\pi k}{N} \right] \right\}} - \frac{\sin \left[\frac{\pi k|m|}{N} \right]}{\sin \left[\frac{\pi k}{N} \right]} \right\} \quad (6)$$

PROPAGATOR IN RING LATTICE WITH ABSORPTION SITE

The occupation probability for the periodic ring lattice with arbitrary K and an absorption site at $n = m$ can be written via the general relation for any $t \geq 1$

$$W_{n_0}(n, t) = Q_{n_0}^{(k)}(n, t) - \sum_{s=1}^t \mathcal{F}_{n_0 \rightarrow m}(s) Q_m^{(k)}(n, t-s) \quad (7)$$

where $Q_r^{(k)}(n, t)$ is the propagator for the periodic ring lattice in the absence of any absorption site. Note that at $t = 0$, by construction $F_{n_0 \rightarrow m}(0) = 0$ when $n_0 \neq m$, and we have $W_{n_0}(n, 0) = Q_{n_0}^{(k)}(n, 0)$.

More explicit expressions can be obtained for the case $K = 2$, that is the nearest-neighbour lattice walk. In the compact odd-mode notation below, and throughout the antipodal shortcut calculation, we take N even. We have that

$$Q_{n_0}^{(1)}(n, t) = \frac{1}{N} \sum_{k=0}^{N-1} \cos \left[\frac{2\pi k(n-n_0)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t, \quad (8)$$

and for $t \geq 1$

$$\mathcal{F}_{n_0}(m, t) = \frac{q}{N} \sum_{k=1}^{N-1} [1 - (-1)^k] \sin \left[\frac{|m-n_0|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right] \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^{t-1}. \quad (9)$$

with $\mathcal{F}_{n_0}(m, 0) = 0$.

Given that in Eq. (9) only odd k terms are non-zero, the first-passage probability can be written in compact form as

$$\mathcal{F}_{n_0}(m, t) = \sum_{k=1}^{\frac{N}{2}} f_k(n_0, m) \alpha_k^{t-1}. \quad (10)$$

where

$$f_k(n_0, m) = \frac{2q}{N} \sin \left[\frac{|m-n_0|\pi(2k-1)}{N} \right] \sin \left[\frac{\pi(2k-1)}{N} \right] \quad (11)$$

and

$$\alpha_k = 1 - q + q \cos \left[\frac{\pi(2k-1)}{N} \right]. \quad (12)$$

The generating function of Eq. (10) in compact form can be written as

$$\tilde{\mathcal{F}}_{n_0}(m, z) = z \sum_{k=1}^{\frac{N}{2}} \frac{f_k(n_0, m)}{1 - z\alpha_k} = \frac{U_{N-|m-n_0|-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) + U_{|m-n_0|-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right)}{U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right)}. \quad (13)$$

where $U_r(y)$ is the Chebyshev polynomial of the second kind.

Also we note that using Eq. (1) one can show that the cumulative absorption probability is

$$\sum_{s=1}^t \mathcal{F}_{n_0}(m, s) = 1 - \frac{1}{N} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|n_0-m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t,$$

which gives the survival probability

$$R_{n_0}(t) = 1 - \sum_{s=0}^t \mathcal{F}_{n_0}(m, s) = \frac{1}{N} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|n_0-m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t.$$

Using the above expressions in Eq. (7) and evaluating the time summation, we obtain

$$\begin{aligned}
W_{n_0}(n, t) &= \frac{1}{N} \sum_{k=1}^{N-1} \cos \left[\frac{2\pi k(n-n_0)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t \\
&+ \frac{1}{N^2} \sum_{k=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|n_0-m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \\
&+ \frac{1}{N^2} \sum_{k=1}^{N-1} \sum_{r=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|n_0-m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right] \cos \left[\frac{2\pi r(n-m)}{N} \right]}{\cos \left[\frac{\pi k}{N} \right] - \cos \left[\frac{2\pi r}{N} \right]} \\
&\times \left\{ \left[1 - q + q \cos \left(\frac{2\pi r}{N} \right) \right]^t - \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \right\}
\end{aligned} \tag{14}$$

Using the identities (4) and (6), we reduce the double summation to single summations as follows

$$\begin{aligned}
&\sum_{k=1}^{N-1} \sum_{r=1}^{N-1} \frac{[1 - (-1)^k] \sin \left[\frac{|n_0-m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right] \cos \left[\frac{2\pi r(n-m)}{N} \right]}{\cos \left[\frac{\pi k}{N} \right] - \cos \left[\frac{2\pi r}{N} \right]} \left\{ \left[1 - q + q \cos \left(\frac{2\pi r}{N} \right) \right]^t - \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \right\} \\
&= -N \sum_{r=1}^{N-1} \cos \left[\frac{2\pi r|n_0-m|}{N} \right] \cos \left[\frac{2\pi r(n-m)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi r}{N} \right) \right]^t \\
&- \sum_{k=1}^{N-1} [1 - (-1)^k] \left\{ \frac{\sin \left[\frac{|n_0-m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} - N \sin \left[\frac{|n_0-m|\pi k}{N} \right] \sin \left[\frac{|n-m|\pi k}{N} \right] \right\} \\
&\times \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \\
&= -N \sum_{r=1}^{N-1} \cos \left[\frac{2\pi r|n_0-m|}{N} \right] \cos \left[\frac{2\pi r(n-m)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi r}{N} \right) \right]^t \\
&- \sum_{k=1}^{N-1} [1 - (-1)^k] \left\{ \frac{\sin \left[\frac{|n_0-m|\pi k}{N} \right] \sin \left[\frac{\pi k}{N} \right]}{1 - \cos \left[\frac{\pi k}{N} \right]} \right. \\
&\left. - N \sin \left[\frac{|n_0-m|\pi k}{N} \right] \sin \left[\frac{|n-m|\pi k}{N} \right] \right\} \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t
\end{aligned} \tag{15}$$

and we have

$$\begin{aligned}
W_{n_0}(n, t) &= \frac{1}{N} \sum_{k=1}^{N-1} \cos \left[\frac{2\pi k(n-n_0)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t \\
&- \frac{1}{N} \sum_{k=1}^{N-1} \cos \left[\frac{2\pi k(n-m)}{N} \right] \cos \left[\frac{2\pi k(n_0-m)}{N} \right] \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t \\
&+ \frac{1}{N} \sum_{k=1}^{N-1} [1 - (-1)^k] \sin \left[\frac{\pi k|n_0-m|}{N} \right] \sin \left[\frac{\pi k|n-m|}{N} \right] \left\{ 1 - q + q \cos \left[\frac{\pi k}{N} \right] \right\}^t.
\end{aligned} \tag{16}$$

The initial condition follows from the complete orthogonality relations. Let

$$a_n \equiv n - m \pmod{N}, \quad b \equiv n_0 - m \pmod{N},$$

with representatives in $1, \dots, N-1$. Then

$$\sum_{k=1}^{N-1} \cos \left(\frac{2\pi k a_n}{N} \right) \cos \left(\frac{2\pi k b}{N} \right) = \frac{N}{2} (\delta_{a_n, b} + \delta_{a_n, N-b}) - 1,$$

and

$$\sum_{k=1}^{N-1} [1 - (-1)^k] \sin\left(\frac{\pi k |a_n|}{N}\right) \sin\left(\frac{\pi k |b|}{N}\right) = \frac{N}{2} (\delta_{a_n, b} + \delta_{a_n, N-b}).$$

Substituting these identities in Eq. (16) gives $W_{n_0}(n, 0) = \delta_{n, n_0}$ for $n, n_0 \neq m$. Furthermore, for the killed propagator W , when either $n = m$ or $n_0 = m$ one has $W_{n_0}(n, t) = 0$, and it is also easy to see that $W_{n_0}(n, t \rightarrow \infty) = 0$, which means that all random walks eventually reach the absorbing site $n = m$.

We can rewrite Eq. (16) as

$$W_{n_0}(n, t) = \sum_{k=1}^{N-1} G_k^{(1)}(n_0, n, m) \left[1 - q + q \cos\left(\frac{2\pi k}{N}\right)\right]^t + \sum_{k=1}^{N-1} G_k^{(2)}(n_0, n, m) \left[1 - q + q \cos\left(\frac{\pi k}{N}\right)\right]^t \quad (17)$$

where m is the absorbing site, and with

$$\begin{aligned} G_k^{(1)}(n_0, n, m) &= \frac{\cos\left[\frac{2\pi k(n-n_0)}{N}\right] - \cos\left[\frac{2\pi k(n-m)}{N}\right] \cos\left[\frac{2\pi k(n_0-m)}{N}\right]}{N} \\ G_k^{(2)}(n_0, n, m) &= \frac{[1 - (-1)^k] \sin\left[\frac{\pi k |n_0-m|}{N}\right] \sin\left[\frac{\pi k |n-m|}{N}\right]}{N} \end{aligned} \quad (18)$$

In terms of generating functions one has

$$\widetilde{W}_{n_0}(n, z) = \sum_{k=1}^{N-1} \frac{G_k^{(1)}(n_0, n, m)}{1 - z [1 - q + q \cos(\frac{2\pi k}{N})]} + \sum_{k=1}^{N-1} \frac{G_k^{(2)}(n_0, n, m)}{1 - z [1 - q + q \cos(\frac{\pi k}{N})]} \quad (19)$$

or the equivalent form

$$\widetilde{W}_{n_0}(n, z) = \sum_{k=1}^{N-1} \frac{G_k^{(1)}(n_0, n, m)}{1 - z \gamma_k} + \sum_{k=1}^{\frac{N}{2}} \frac{g_k^{(2)}(n_0, n, m)}{1 - z \alpha_k}, \quad (20)$$

and its time-dependent version

$$W_{n_0}(n, t) = \sum_{k=1}^{N-1} G_k^{(1)}(n_0, n, m) \gamma_k^t + \sum_{k=1}^{\frac{N}{2}} g_k^{(2)}(n_0, n, m) \alpha_k^t, \quad (21)$$

where α_k is defined in Eq. (12),

$$\gamma_k = 1 - q + q \cos\left(\frac{2\pi k}{N}\right), \quad (22)$$

and

$$g_k^{(2)}(n_0, n, m) = \frac{2}{N} \sin\left[\frac{\pi(2k-1)|n_0-m|}{N}\right] \sin\left[\frac{\pi(2k-1)|n-m|}{N}\right]. \quad (23)$$

The associated expressions in terms of Chebyshev polynomials of the first and second kind, respectively, $T_r(y)$ and $U_r(y)$, are written with residues modulo N represented in $0, \dots, N-1$:

$$\begin{aligned} \sum_{k=1}^{N-1} \frac{G_k^{(1)}(n_0, n, m)}{1 - z [1 - q + q \cos(\frac{2\pi k}{N})]} &= \left\{ 2 T_{N-|n-n_0|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1\right]\right) + 2 T_{|n-n_0|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1\right]\right) \right. \\ &\quad - T_{N-|n_0-m|-|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1\right]\right) - T_{|n_0-m|-|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1\right]\right) \\ &\quad \left. - T_{N-[|n_0-m|+|n-m| \pmod{N}]} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1\right]\right) - T_{|n_0-m|+|n-m| \pmod{N}} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1\right]\right) \right\} \\ &\times \left[2zq \left\{ \left[1 + \frac{1}{q} \left(\frac{1}{z} - 1\right)\right]^2 - 1 \right\} U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1\right]\right) \right]^{-1}, \end{aligned} \quad (24)$$

and

$$\begin{aligned} \sum_{k=1}^{N-1} \frac{G_k^{(2)}(n_0, n, m)}{1 - z \left[1 - q + q \cos\left(\frac{\pi k}{N}\right) \right]} &= \left\{ T_{N-|n_0-m|-|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) + T_{N-|N-|n_0-m|-|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right. \\ &\quad \left. - T_{|N-|n_0-m|-|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - T_{|n_0-m|-|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right\} \\ &\quad \times \left[2zq \left\{ \left[1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right) \right]^2 - 1 \right\} U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right]^{-1}, \end{aligned} \quad (25)$$

When $n = n_0$ Eq. (17) reduces to

$$\begin{aligned} W_n(n, t) &= \frac{1}{N} \sum_{k=1}^{N-1} \left\{ \left(1 - \cos^2 \left[\frac{2\pi k(n-m)}{N} \right] \right) \left[1 - q + q \cos \left(\frac{2\pi k}{N} \right) \right]^t \right. \\ &\quad \left. + [1 - (-1)^k] \sin^2 \left[\frac{\pi k|n-m|}{N} \right] \left[1 - q + q \cos \left(\frac{\pi k}{N} \right) \right]^t \right\}, \end{aligned} \quad (26)$$

whose generating function is given by the compact expression

$$\begin{aligned} \widetilde{W}_n(n, z) &= \left\{ 2T_N \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) + T_{N-|N-2|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - T_{|N-2|n-m|} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right. \\ &\quad \left. - T_{N-[2|n-m| \pmod{N}]} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - T_{[2|n-m| \pmod{N}]} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right\} \\ &\quad \left\{ 2zq \left\{ \left[1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right) \right]^2 - 1 \right\} U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right\}^{-1} \end{aligned} \quad (27)$$

Note that using Eqs. (19) with $n = n_0$ and (27) when $|n-m| = N/2$ with N even, one can show setting $z = 1$ the following identities

$$\sum_{k=1}^{N-1} \frac{1 - (-1)^k}{1 - \cos\left(\frac{\pi k}{N}\right)} = \frac{N^2}{2}, \quad \sum_{k=1}^{N-1} \frac{1}{1 - \cos\left(\frac{\pi k}{N}\right)} = \frac{N^2 - 1}{3}, \quad \sum_{k=1}^{N-1} \frac{(-1)^k}{1 - \cos\left(\frac{\pi k}{N}\right)} = -\frac{N^2 + 2}{6}.$$

For the special case $|n-m| = N/2$ with N even Eq. (27) becomes

$$\widetilde{W}_n(n, z) = \frac{T_N \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) - 1}{zq \left\{ \left[1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right) \right]^2 - 1 \right\} U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right)} = \frac{2 \left[U_{\frac{N}{2}-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right) \right]^2}{zq U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1 \right] \right)} \quad (28)$$

PROPAGATOR WITH ONE SHORTCUT AND ONE ABSORBING SITE

We start from the general propagator with a directed shortcut from $n = u$ to site $n = v$. The shortcut reallocates the transition probability $\beta(1-q)$ from the lazy self-loop at u to the directed edge $u \rightarrow v$. Thus the u -row remains stochastic. We have

$$\widetilde{S}_{n_0}(n, z) = \widetilde{P}_{n_0}(n, z) - \frac{z\beta(1-q)\widetilde{P}_{n_0}(u, z) \left[\widetilde{P}_u(n, z) - \widetilde{P}_v(n, z) \right]}{1 + z\beta(1-q) \left[\widetilde{P}_u(u, z) - \widetilde{P}_v(u, z) \right]}, \quad (29)$$

where $\widetilde{P}_r(n, z)$ is the ring propagator generating function without shortcuts. If the ring also has an absorbing site, the modularity of the defect technique allows to replace the (probability conserving) propagator $\widetilde{P}_r(n, z)$ with the

propagator $\widetilde{W}_r(n, z)$ in Eq. (17)

$$\widetilde{S}_{n_0}(n, z) = \widetilde{W}_{n_0}(n, z) - \frac{z\beta(1-q)\widetilde{W}_{n_0}(u, z) [\widetilde{W}_u(n, z) - \widetilde{W}_v(n, z)]}{1 + z\beta(1-q) [\widetilde{W}_u(u, z) - \widetilde{W}_v(u, z)]}. \quad (30)$$

When the absorbing site coincides with the shortcut site, i.e. $m = v$, $\widetilde{W}_v(n, z) = 0$ and Eq. (30) simplifies further to

$$\widetilde{S}_{n_0}(n, z) = \widetilde{W}_{n_0}(n, z) - \frac{z\beta(1-q)\widetilde{W}_{n_0}(u, z)\widetilde{W}_u(n, z)}{1 + z\beta(1-q)\widetilde{W}_u(u, z)}. \quad (31)$$

After summing Eq. (31) over n to obtain the survival probability $\widetilde{S}_{n_0}(z) = [1 - \widetilde{F}_{n_0}(v, z)] / (1 - z)$, one expresses the generating function of the first-passage probability to site v as

$$\widetilde{F}_{n_0}(v, z) = \widetilde{F}_{n_0}(v, z) + \frac{z\beta(1-q)\widetilde{W}_{n_0}(u, z) [1 - \widetilde{F}_u(v, z)]}{1 + z\beta(1-q)\widetilde{W}_u(u, z)}, \quad (32)$$

or more explicitly as

$$\widetilde{F}_{n_0}(v, z) = \widetilde{F}_{n_0}(v, z) + qz\widetilde{W}_{n_0}(u, z) [1 - \widetilde{F}_u(v, z)] \widetilde{H}_{N,|u-v|}(z), \quad (33)$$

with $\widetilde{W}_{n_0}(u, z)$ and $\widetilde{F}_r(m, z)$ defined, respectively, in Eqs. (19) and (13), and

$$\widetilde{H}_{N,|u-v|}(z) = \frac{\beta(1-q)}{q} \frac{1}{1 + z\beta(1-q)\widetilde{W}_u(u, z)}. \quad (34)$$

We now consider the symmetric case $|u - v| = N/2$ with N even, and use $\widetilde{W}_u(u, z)$ from Eq. (28) in Eq. (34), which allows us to simplify Eq. (34) to

$$\widetilde{H}_{N,N/2}(z) = \frac{1}{\frac{q}{\beta(1-q)} + \phi(z)}, \quad (35)$$

where

$$\phi(z) = \frac{2U_{N/2-1}^2 \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1\right]\right)}{U_{N-1} \left(1 + \frac{1}{q} \left[\frac{1}{z} - 1\right]\right)}. \quad (36)$$

whose time-dependence is $H_{N,N/2}(0) = \beta(1-q)/q$ and for $t \geq 1$ is given by the complex integral after the change of integration variable $z = 1/s$

$$H_{N,N/2}(t) = \frac{1}{2\pi i} \oint_{|s|=R} \frac{s^{t-1}}{\frac{q}{\beta(1-q)} + \phi\left(\frac{1}{s}\right)} ds, \quad (37)$$

where R is chosen so that the relevant poles are enclosed. The denominator in the integrand of Eq. (37) has real zeros s_k , which are the intersections of $\phi(1/s)$ with the horizontal line $-q/[\beta(1-q)]$. These intersections are shown graphically in Fig. 1.

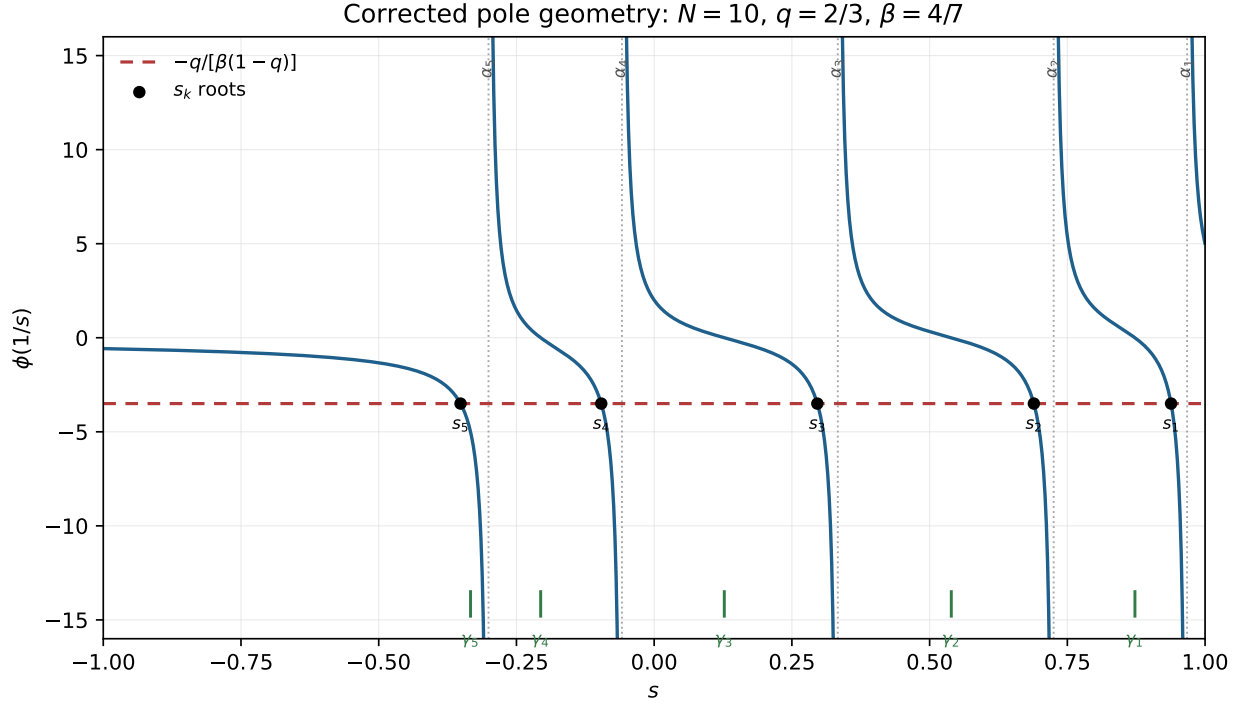


FIG. 1. Graphical representation of the poles of Eq. (35) for the case $N = 10$, $q = 2/3$, and $\beta = 4/7$. The poles are the values of s where the horizontal line $-q/[\beta(1-q)]$ intercepts $\phi(1/s)$. The asymptotes of $\phi(1/s)$ are given by the α_k , and the paired tick marks indicate the corresponding γ_k . This figure is a visualization of the H -poles; the finite-time one-peak/two-peak transition is a property of the full first-passage distribution.

It is useful to rewrite the same corrected H in a form tied directly to its denominator. Let $N = 2L$,

$$a = \frac{q}{\beta(1-q)}, \quad y = 1 + \frac{1/z - 1}{q}.$$

Then

$$\tilde{H}_{N,N/2}(z) = \frac{T_L(y)}{aT_L(y) + U_{L-1}(y)}. \quad (38)$$

If $D(y) = aT_L(y) + U_{L-1}(y)$, $D(y_k) = 0$, and $s_k = 1 - q + qy_k$, then the positive-time part of H can be written, for simple roots, as

$$H_{N,N/2}(t) = \sum_k c_k s_k^{t-1}, \quad c_k = \frac{qT_L(y_k)}{D'(y_k)}, \quad t \geq 1. \quad (39)$$

This residue form keeps the $H(0) = \beta(1-q)/q$ contribution separate from the positive-time pole expansion.

The time dependence of Eq. (33) for $|u - v| = N/2$ is therefore given by the time-convoluted expression for $t \geq 1$

$$F_{n_0}(v, t) = \mathcal{F}_{n_0}(v, t) + q \sum_{t'=0}^{t-1} W_{n_0}(u, t-1-t') \sum_{t''=0}^{t'} [\delta_{t', t''} - \mathcal{F}_u(v, t' - t'')] H_{N,N/2}(t'') \quad (40)$$

and $F_{n_0}(v, 0) = 0$.

The same result can be written more compactly at the generating-function level. Let ρ denote the folded distance

of the starting site from the absorbing target in the antipodal case. Using

$$\tilde{\mathcal{F}}_\rho(0, z) = \frac{T_{L-\rho}(y)}{T_L(y)}, \quad (41)$$

$$\tilde{W}_\rho(L, z) = \frac{U_{\rho-1}(y)}{zq T_L(y)}, \quad (42)$$

$$1 - \tilde{\mathcal{F}}_L(0, z) = \frac{T_L(y) - 1}{T_L(y)}, \quad (43)$$

$$\tilde{H}_{N, N/2}(z) = \frac{T_L(y)}{aT_L(y) + U_{L-1}(y)}, \quad (44)$$

Eq. (33) reduces to

$$\begin{aligned} \tilde{F}_\rho(0, z) &= \frac{T_{L-\rho}(y)}{T_L(y)} + \frac{U_{\rho-1}(y)[T_L(y) - 1]}{T_L(y)[aT_L(y) + U_{L-1}(y)]} \\ &= \frac{aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)}{aT_L(y) + U_{L-1}(y)}. \end{aligned} \quad (45)$$

The second equality uses

$$T_L(y)U_{L-\rho-1}(y) = T_{L-\rho}(y)U_{L-1}(y) - U_{\rho-1}(y).$$

Here and below $U_{-1}(y) = 0$, which covers the endpoint case $\rho = L$. This cancellation should be read as a cancellation in the fully collected rational function, not as an attempt to determine poles from a numerator alone. Before reduction, Eq. (45) has the common denominator $T_L(y)\{aT_L(y) + U_{L-1}(y)\}$, so the zeros of $T_L(y)$ are candidate poles at that unreduced stage. The numerator over this common denominator is

$$M_\rho(y) = T_{L-\rho}(y)\{aT_L(y) + U_{L-1}(y)\} + U_{\rho-1}(y)\{T_L(y) - 1\}.$$

This $M_\rho(y)$ is a polynomial expression in y , with no hidden $1/T_L(y)$ factor. The Chebyshev identity above then gives

$$M_\rho(y) = T_L(y)[aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)].$$

Thus $T_L(y)$ is a common analytic factor of the collected numerator and denominator. It can be removed only after this collection step. The expanded-route check in the Appendix shows the same cancellation explicitly, including the diagonal limits in the time-domain partial fractions.

The tail implication is therefore best read from the collected expression, not from the uncollected term-by-term convolution. The apparent α_ℓ -modes that arise in the time-domain hard route correspond to the zeros of $T_L(y)$. They are removable singularities of the collected expression after the common $T_L(y)$ factor is removed. Hence the long-time behaviour is determined by the zeros of

$$D(y) = aT_L(y) + U_{L-1}(y).$$

It is not determined by the zeros of $T_L(y)$. If $D(y_k) = 0$ and $s_k = 1 - q + qy_k$, then for simple roots

$$F_\rho(0, t) = \sum_k B_{\rho k} s_k^{t-1}, \quad B_{\rho k} = \frac{q[aT_{L-\rho}(y_k) + U_{\rho-1}(y_k) + U_{L-\rho-1}(y_k)]}{D'(y_k)}. \quad (46)$$

Therefore, ordering the roots so that s_1 is the dominant positive pole and assuming $B_{\rho 1} \neq 0$, the corrected long-time dependence is

$$F_\rho(0, t) \sim B_{\rho 1} s_1^{t-1}, \quad t \rightarrow \infty.$$

Equivalently, the exponential decay rate is $-\log s_1$. The dependence on the shortcut strength enters through

$$a = \frac{q}{\beta(1-q)}$$

in the denominator $D(y) = aT_L(y) + U_{L-1}(y)$, and hence through the movement of the dominant root $s_1 = 1 - q + qy_1$. Thus the corrected shortcut calculation changes the tail by moving the dominant pole; it does not produce a generic $t\alpha_1^t$ long-time correction. The finite-time one-peak/two-peak transition is therefore a shape question for the full first-passage distribution, not a criterion obtained from the auxiliary H -pole plot alone. In particular, for finite positive β in the corrected pole equation, the proposed crossing $s_1 = \gamma_1 = 1 - q + q\cos(2\pi/N)$ does not occur; the dominant corrected pole satisfies $\gamma_1 < s_1 < \alpha_1$. The deterministic finite-matrix numerical audit used to check these statements is recorded in Appendix C, including the direct shortcut matrix benchmark and the five-group check of Eq. (57).

APPENDIX A: DIRECT TIME-DOMAIN ROUTE

This appendix carries out the corrected calculation in the same time-convolution route as the long displayed expansion. The goal is to make clear which terms arise before simplification, and why the collected answer is the compact closed form in Eq. (45). Throughout this appendix

$$N = 2L, \quad L \geq 2, \quad 0 < q < 1, \quad 0 < \beta \leq 1,$$

the absorbing target is $v = 0$, the shortcut source is $u = L$, and $1 \leq \rho \leq L$ is the folded distance from the initial site to v . We also use

$$a = \frac{q}{\beta(1-q)}, \quad y = 1 + \frac{1/z - 1}{q}, \quad U_{-1}(y) = 0.$$

A.1 Corrected convolution

The corrected generating-function starting point is Eq. (33),

$$\tilde{F}_{n_0}(v, z) = \tilde{\mathcal{F}}_{n_0}(v, z) + qz \tilde{W}_{n_0}(u, z) \left[1 - \tilde{\mathcal{F}}_u(v, z) \right] \tilde{H}_{N, N/2}(z). \quad (47)$$

Equivalently, for $t \geq 1$,

$$F_{n_0}(v, t) = \mathcal{F}_{n_0}(v, t) + q \sum_{t'=0}^{t-1} W_{n_0}(u, t-1-t') \sum_{t''=0}^{t'} [\delta_{t', t''} - \mathcal{F}_u(v, t' - t'')] H_{N, N/2}(t''). \quad (48)$$

This is the route expanded below.

A.2 Mode inputs

For $N = 2L$, the no-shortcut first-passage probability is

$$\mathcal{F}_{n_0}(v, t) = \sum_{\ell=1}^L f_{\ell}(n_0, v) \alpha_{\ell}^{t-1}, \quad t \geq 1, \quad (49)$$

where f_{ℓ} is defined in Eq. (11) with the odd index $2\ell - 1$, and

$$\alpha_{\ell} = 1 - q + q \cos \frac{(2\ell - 1)\pi}{N}.$$

From Eq. (17), evaluated at the shortcut source u , we write

$$W_{n_0}(u, t) = \sum_{r=1}^{N-1} G_r^{(1)}(n_0, u, v) \gamma_r^t + \sum_{r=1}^L g_r^{(2)}(n_0, u, v) \alpha_r^t, \quad (50)$$

where

$$\gamma_r = 1 - q + q \cos \frac{2\pi r}{N}, \quad g_r^{(2)}(n_0, u, v) = G_{2r-1}^{(2)}(n_0, u, v).$$

The even $G^{(2)}$ -terms vanish because of the factor $1 - (-1)^k$.

For the antipodal shortcut,

$$\tilde{H}_{N, N/2}(z) = \frac{T_L(y)}{D(y)}, \quad D(y) = aT_L(y) + U_{L-1}(y). \quad (51)$$

The $H(0)$ term must be kept separate:

$$h_0 = H_{N, N/2}(0) = \frac{\beta(1-q)}{q} = \frac{1}{a}.$$

If $D(y_j) = 0$ and $s_j = 1 - q + qy_j$, then the positive-time part is

$$H_{N, N/2}(t) = \sum_j c_j s_j^{t-1}, \quad c_j = \frac{qT_L(y_j)}{D'(y_j)}, \quad t \geq 1. \quad (52)$$

A.3 Convolution kernels

The δ -term in Eq. (48) gives a two-factor convolution. The $H(0)$ part is simply

$$A_t^{(0)}(x) = x^{t-1}.$$

For a positive-time H -pole,

$$A_t^{(+)}(x; s) = \sum_{c=1}^{t-1} x^{t-1-c} s^{c-1} = \frac{s^{t-1} - x^{t-1}}{s - x}. \quad (53)$$

The coincident case $x = s$ is understood as the limit

$$A_t^{(+)}(s; s) = (t-1)s^{t-2}.$$

The negative $\mathcal{F}_u(v, \cdot)$ -part gives a three-factor convolution. With $b \geq 1$, the $H(0)$ contribution is

$$B_t^{(0)}(x, y) = \sum_{\substack{a+b=t-1 \\ b \geq 1}} x^a y^{b-1} = \frac{x^{t-1} - y^{t-1}}{x - y}. \quad (54)$$

The positive-time H -pole contribution is

$$\begin{aligned} B_t^{(+)}(x, y; s) &= \sum_{\substack{a+b+c=t-1 \\ b \geq 1, c \geq 1}} x^a y^{b-1} s^{c-1} \\ &= \text{Coeff}_{\xi^{t-3}} \frac{1}{(1 - \xi x)(1 - \xi y)(1 - \xi s)}. \end{aligned} \quad (55)$$

When x, y, s are distinct,

$$B_t^{(+)}(x, y; s) = \frac{x^{t-1}}{(x-y)(x-s)} + \frac{y^{t-1}}{(y-x)(y-s)} + \frac{s^{t-1}}{(s-x)(s-y)}. \quad (56)$$

Coincident cases are obtained by taking limits. This is the point at which an uncollected term-by-term expansion can temporarily produce factors such as $t\alpha_\ell^t$; those factors are not tail terms unless they survive after the complete expression is recombined.

A.4 Corrected Eq. (41)-style expansion

Substituting Eqs. (49), (50), and (52) into Eq. (48) gives the corrected long expansion. To keep the direct comparison with the previous long Eq. (41)-style calculation transparent, all five convolution groups are written out

below; no A_t - or B_t -kernel shorthand is used in the displayed result.

$$\begin{aligned}
F_{n_0}(v, t) = & \sum_{\ell=1}^L f_{\ell}(n_0, v) \alpha_{\ell}^{t-1} \\
& + q h_0 \left[\sum_{r=1}^{N-1} G_r^{(1)}(n_0, u, v) \gamma_r^{t-1} + \sum_{r=1}^L g_r^{(2)}(n_0, u, v) \alpha_r^{t-1} \right] \\
& + q \sum_j c_j \left[\sum_{r=1}^{N-1} G_r^{(1)}(n_0, u, v) \frac{s_j^{t-1} - \gamma_r^{t-1}}{s_j - \gamma_r} + \sum_{r=1}^L g_r^{(2)}(n_0, u, v) \frac{s_j^{t-1} - \alpha_r^{t-1}}{s_j - \alpha_r} \right] \\
& - q h_0 \sum_{\ell=1}^L f_{\ell}(u, v) \left[\sum_{r=1}^{N-1} G_r^{(1)}(n_0, u, v) \frac{\gamma_r^{t-1} - \alpha_{\ell}^{t-1}}{\gamma_r - \alpha_{\ell}} \right. \\
& \quad \left. + \sum_{\substack{r=1 \\ r \neq \ell}}^L g_r^{(2)}(n_0, u, v) \frac{\alpha_r^{t-1} - \alpha_{\ell}^{t-1}}{\alpha_r - \alpha_{\ell}} + g_{\ell}^{(2)}(n_0, u, v) (t-1) \alpha_{\ell}^{t-2} \right] \\
& - q \sum_j c_j \sum_{\ell=1}^L f_{\ell}(u, v) \left[\sum_{r=1}^{N-1} G_r^{(1)}(n_0, u, v) \left(\frac{\gamma_r^{t-1}}{(\gamma_r - \alpha_{\ell})(\gamma_r - s_j)} + \frac{\alpha_{\ell}^{t-1}}{(\alpha_{\ell} - \gamma_r)(\alpha_{\ell} - s_j)} + \frac{s_j^{t-1}}{(s_j - \gamma_r)(s_j - \alpha_{\ell})} \right) \right. \\
& \quad + \sum_{\substack{r=1 \\ r \neq \ell}}^L g_r^{(2)}(n_0, u, v) \left(\frac{\alpha_r^{t-1}}{(\alpha_r - \alpha_{\ell})(\alpha_r - s_j)} + \frac{\alpha_{\ell}^{t-1}}{(\alpha_{\ell} - \alpha_r)(\alpha_{\ell} - s_j)} + \frac{s_j^{t-1}}{(s_j - \alpha_r)(s_j - \alpha_{\ell})} \right) \\
& \quad \left. + g_{\ell}^{(2)}(n_0, u, v) \left(\frac{s_j^{t-1} - \alpha_{\ell}^{t-1}}{(s_j - \alpha_{\ell})^2} - \frac{(t-1) \alpha_{\ell}^{t-2}}{s_j - \alpha_{\ell}} \right) \right]. \tag{57}
\end{aligned}$$

Equation (57) is the corrected direct-route replacement for the long Eq. (41)-style expression. The baseline first-passage term targets v , the global sign follows the positive sign in Eq. (47), and the $g^{(2)}$ -modes are paired with α_r . The final term in the last bracket is the $r = \ell$ diagonal limit of the $(\alpha_r, \alpha_{\ell}, s_j)$ triple convolution. Thus the fully expanded hard-route form really does contain explicit α_{ℓ} -powers and $t\alpha_{\ell}$ -type factors. The point of the later recombination is not that these intermediate terms are absent, but that their apparent $T_L(y) = 0$ poles cancel after all five convolution groups are collected.

How Eq. (57) is generated, and where the cancellation occurs. The correspondence with the previous long Eq. (41)-style expression is term-by-term, but not by a mere change of notation. The first term of the previous long expression corresponds to the first line of Eq. (57); the target in the baseline term is v , so the coefficient is $f_{\ell}(n_0, v)$, not $f_{\ell}(n_0, u)$. The first large s -dependent brace in the previous long expression corresponds to the second and third lines of Eq. (57): the $H(0) = h_0$ contribution has to be split off, giving the $q h_0 W$ term, and only the positive-time part of H gives the difference quotients $(s_j^{t-1} - x^{t-1})/(s_j - x)$. The previous $G^{(1)} f_{\ell}(u, v)$ terms correspond to the $G^{(1)}$ -parts in the fourth and fifth lines, written as a two-mode difference quotient and a three-mode partial fraction. The previous $g^{(2)} f_{\ell}(u, v)$ terms correspond to the $g^{(2)}$ -parts in the fourth and fifth lines; here $g_r^{(2)}$ is paired with the α_r -mode. Any visible $t\alpha_{\ell}^t$ term is a coincident-limit case of the written-out partial fractions, not a final tail term.

The six term-level correspondences are the following. To keep the displays readable, only in this paragraph write

$$G_r = G_r^{(1)}(n_0, u, v), \quad g_r = g_r^{(2)}(n_0, u, v), \quad f_{\ell}^{uv} = f_{\ell}(u, v), \quad L = N/2.$$

1. The no-shortcut baseline term is

$$\begin{aligned}
\text{previous piece:} & \quad \sum_{\ell=1}^L f_{\ell}(n_0, u) \alpha_{\ell}^{t-1}, \\
\text{corrected piece:} & \quad \sum_{\ell=1}^L f_{\ell}(n_0, v) \alpha_{\ell}^{t-1}.
\end{aligned}$$

This is the first line of Eq. (57); the target of the no-shortcut first passage is v .

2. The first large s -dependent W -brace is replaced by the two pieces $qW * \delta * h_0 \delta_0$ and $qW * \delta * H_+$:

$$\begin{aligned} \text{previous piece:} \quad & -q \sum_{k=1}^M \left[\sum_{r=1}^{N-1} \frac{c_k G_r}{s_k(s_k - \gamma_r)} (s_k^t - \gamma_r^t) \right. \\ & \quad \left. + \sum_{\ell=1}^L \frac{c_k g_\ell}{s_k(s_k - \alpha_\ell)} (s_k^t - \alpha_\ell^t) \right], \\ \text{corrected pieces:} \quad & qh_0 \left[\sum_{r=1}^{N-1} G_r \gamma_r^{t-1} + \sum_{r=1}^L g_r \alpha_r^{t-1} \right] \\ & + q \sum_j c_j \left[\sum_{r=1}^{N-1} G_r \frac{s_j^{t-1} - \gamma_r^{t-1}}{s_j - \gamma_r} + \sum_{r=1}^L g_r \frac{s_j^{t-1} - \alpha_r^{t-1}}{s_j - \alpha_r} \right]. \end{aligned}$$

These are the second and third lines of Eq. (57). The sign changes because the corrected route uses $+qW * (\delta - \mathcal{F}_u) * H$, and $H(0) = h_0$ is no longer folded into the positive-time pole expansion.

3. The power shift in the old s -dependent factors is exactly the $H(0)$ separation:

$$\begin{aligned} \text{old kernel if the positive-time pole is used at time zero:} \quad & \sum_{c=0}^{t-1} x^{t-1-c} s^{c-1} = \frac{s^t - x^t}{s(s-x)}, \\ \text{corrected positive-time sum:} \quad & \sum_{c=1}^{t-1} x^{t-1-c} s^{c-1} = \frac{s^{t-1} - x^{t-1}}{s-x}. \end{aligned}$$

This is why the corrected Eq. (57) uses the $(s^{t-1} - x^{t-1})/(s-x)$ difference quotient rather than the old-looking $(s^t - x^t)/(s(s-x))$ factors.

4. The $G^{(1)} f_\ell(u, v)$ triple-convolution terms are

$$\begin{aligned} \text{previous piece:} \quad & q \sum_{k=1}^M \sum_{r=1}^{N-1} \sum_{\ell=1}^L \frac{c_k G_r f_\ell^{uv}}{s_k - \alpha_\ell} \left[\frac{s_k^t - \gamma_r^t}{\alpha_\ell(s_k - \gamma_r)} - \frac{\alpha_\ell^t - \gamma_r^t}{s_k(\alpha_\ell - \gamma_r)} \right], \\ \text{corrected pieces:} \quad & -qh_0 \sum_{\ell=1}^L f_\ell^{uv} \sum_{r=1}^{N-1} G_r \frac{\gamma_r^{t-1} - \alpha_\ell^{t-1}}{\gamma_r - \alpha_\ell} \\ & - q \sum_j c_j \sum_{\ell=1}^L f_\ell^{uv} \sum_{r=1}^{N-1} G_r \left(\frac{\gamma_r^{t-1}}{(\gamma_r - \alpha_\ell)(\gamma_r - s_j)} \right. \\ & \quad \left. + \frac{\alpha_\ell^{t-1}}{(\alpha_\ell - \gamma_r)(\alpha_\ell - s_j)} + \frac{s_j^{t-1}}{(s_j - \gamma_r)(s_j - \alpha_\ell)} \right). \end{aligned}$$

These are the $G^{(1)}$ -parts of the fourth and fifth lines of Eq. (57).

5. The off-diagonal $g^{(2)} f_\ell(u, v)$ terms are

$$\begin{aligned} \text{previous piece:} \quad & q \sum_{k=1}^M \sum_{\substack{r, \ell=1 \\ r \neq \ell}}^L \frac{c_k g_r f_\ell^{uv}}{s_k \alpha_\ell (s_k - \alpha_\ell)} \left[s_k \frac{s_k^t - \gamma_r^t}{s_k - \gamma_r} - \alpha_\ell \frac{\alpha_\ell^t - \gamma_r^t}{\alpha_\ell - \gamma_r} \right], \\ \text{corrected pieces:} \quad & -qh_0 \sum_{\ell=1}^L f_\ell^{uv} \sum_{\substack{r=1 \\ r \neq \ell}}^L g_r \frac{\alpha_r^{t-1} - \alpha_\ell^{t-1}}{\alpha_r - \alpha_\ell} \\ & - q \sum_j c_j \sum_{\ell=1}^L f_\ell^{uv} \sum_{\substack{r=1 \\ r \neq \ell}}^L g_r \left(\frac{\alpha_r^{t-1}}{(\alpha_r - \alpha_\ell)(\alpha_r - s_j)} \right. \\ & \quad \left. + \frac{\alpha_\ell^{t-1}}{(\alpha_\ell - \alpha_r)(\alpha_\ell - s_j)} + \frac{s_j^{t-1}}{(s_j - \alpha_r)(s_j - \alpha_\ell)} \right). \end{aligned}$$

The corrected g_r -mode is paired with α_r , not with γ_r .

6. The visible diagonal $t\alpha_\ell^t$ term is the $r = \ell$ coincident-limit case of the same corrected kernels:

$$\begin{aligned} \text{previous diagonal piece:} & \quad -q \sum_{k=1}^M \sum_{\ell=1}^L \frac{c_k g_\ell f_\ell^{uv}}{s_k \alpha_\ell (s_k - \alpha_\ell)} \alpha_\ell^t t, \\ \text{corrected diagonal pieces:} & \quad -q h_0 \sum_{\ell=1}^L f_\ell^{uv} g_\ell (t-1) \alpha_\ell^{t-2} \\ & \quad -q \sum_j c_j \sum_{\ell=1}^L f_\ell^{uv} g_\ell \left[\frac{s_j^{t-1} - \alpha_\ell^{t-1}}{(s_j - \alpha_\ell)^2} - \frac{(t-1)\alpha_\ell^{t-2}}{s_j - \alpha_\ell} \right], \end{aligned}$$

where the square bracket is the same as

$$\sum_{c=1}^{t-2} (t-1-c) \alpha_\ell^{t-2-c} s_j^{c-1}.$$

Thus the $t\alpha_\ell^t$ -type factor is only a diagonal-limit term in the uncollected expansion. It is not a final tail term unless the corresponding α_ℓ -pole survives the complete recombination, which it does not.

Thus the long Eq. (57) is obtained by following the term-by-term route and keeping the partial fractions separated. Write

$$H(t) = h_0 \delta_{t,0} + H_+(t), \quad H_+(t) = \sum_j c_j s_j^{t-1} \quad (t \geq 1).$$

Then the five groups in Eq. (57) come from the five pieces

$$\begin{aligned} \mathcal{F}_{n_0}(v, \cdot) & \longrightarrow \text{first line of Eq. (57),} \\ q W_{n_0}(u, \cdot) * \delta * h_0 \delta_0 & \longrightarrow \text{second line,} \\ q W_{n_0}(u, \cdot) * \delta * H_+ & \longrightarrow \text{third line,} \\ -q W_{n_0}(u, \cdot) * \mathcal{F}_u(v, \cdot) * h_0 \delta_0 & \longrightarrow \text{fourth line,} \\ -q W_{n_0}(u, \cdot) * \mathcal{F}_u(v, \cdot) * H_+ & \longrightarrow \text{fifth line.} \end{aligned}$$

The sign pattern comes directly from the corrected shortcut contribution $+q W * (\delta - \mathcal{F}_u) * H$: the two δ -pieces are positive, and the two $\mathcal{F}_u$ -pieces are negative. This is the same mechanical route that produces a long Eq. (41)-style expression: insert the modal forms for $\mathcal{F}$, W , and H , do the finite time sums, and leave the answer as separated partial fractions. At that stage the α_ℓ -dependent pieces are visible because the no-shortcut first-passage and killed-propagator modes contain α_ℓ . Reading the tail directly from this uncollected list is premature.

The cancellation is not a cancellation inside a single displayed line of Eq. (57). It occurs when all five groups are recombined before the pole set is read. In generating-function form those same five groups add to

$$\frac{T_{L-\rho}(y)}{T_L(y)} + \frac{U_{\rho-1}(y) [T_L(y) - 1]}{T_L(y) D(y)}.$$

Putting this over the common denominator $T_L(y) D(y)$ gives Eq. (60). The numerator is then

$$a T_{L-\rho}(y) T_L(y) + T_{L-\rho}(y) U_{L-1}(y) + U_{\rho-1}(y) T_L(y) - U_{\rho-1}(y),$$

and the Chebyshev identity factors this numerator as

$$T_L(y) [a T_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)].$$

This displayed $T_L(y)$ factor cancels the apparent $T_L(y)$ in the common denominator. Therefore the α_ℓ -modes that appear while deriving Eq. (57) are removable intermediate modes, not final tail poles.

A.5 Generating-function recombination of Eq. (57)

One can also see directly that Eq. (57) recombines into the generating-function route used below. Use the transform

$$\mathcal{Z}[X_t](z) = \sum_{t \geq 1} X_t z^t.$$

The convolution kernels in Appendix A.3 satisfy

$$\sum_{t \geq 1} A_t^{(0)}(x) z^t = \frac{z}{1 - zx}, \quad \sum_{t \geq 1} A_t^{(+)}(x; s) z^t = \frac{z^2}{(1 - zx)(1 - zs)},$$

and

$$\sum_{t \geq 1} B_t^{(0)}(x, y) z^t = \frac{z^2}{(1 - zx)(1 - zy)}, \quad \sum_{t \geq 1} B_t^{(+)}(x, y; s) z^t = \frac{z^3}{(1 - zx)(1 - zy)(1 - zs)}.$$

The same formulae hold in coincident cases by taking limits.

With the mode inputs above, write

$$\begin{aligned} \widetilde{W}_{n_0}(u, z) &= \sum_{r=1}^{N-1} \frac{G_r^{(1)}(n_0, u, v)}{1 - z\gamma_r} + \sum_{r=1}^L \frac{g_r^{(2)}(n_0, u, v)}{1 - z\alpha_r}, \\ \widetilde{\mathcal{F}}_u(v, z) &= \sum_{\ell=1}^L \frac{f_\ell(u, v)z}{1 - z\alpha_\ell}, \quad \widetilde{H}(z) = h_0 + \widetilde{H}_+(z), \quad \widetilde{H}_+(z) = \sum_j \frac{c_j z}{1 - zs_j}. \end{aligned}$$

Then the two positive δ -pieces in Eq. (57) transform as

$$qh_0 z \widetilde{W}_{n_0}(u, z) + qz \widetilde{W}_{n_0}(u, z) \widetilde{H}_+(z) = qz \widetilde{W}_{n_0}(u, z) \widetilde{H}(z).$$

Similarly, the two negative $\mathcal{F}_u$ -pieces transform as

$$-qh_0 z \widetilde{W}_{n_0}(u, z) \widetilde{\mathcal{F}}_u(v, z) - qz \widetilde{W}_{n_0}(u, z) \widetilde{\mathcal{F}}_u(v, z) \widetilde{H}_+(z) = -qz \widetilde{W}_{n_0}(u, z) \widetilde{\mathcal{F}}_u(v, z) \widetilde{H}(z).$$

Adding also the no-shortcut baseline term gives

$$\widetilde{F}_{n_0}(v, z) = \widetilde{\mathcal{F}}_{n_0}(v, z) + qz \widetilde{W}_{n_0}(u, z) \left[1 - \widetilde{\mathcal{F}}_u(v, z) \right] \widetilde{H}(z).$$

This is Eq. (58). Thus Appendix B is the generating-function recombination of the fully expanded Eq. (57), not a separate shortcut calculation.

APPENDIX B: RECOMBINATION TO THE CLOSED FORM

The same route can now be collected at the generating-function level. Let

$$D(y) = aT_L(y) + U_{L-1}(y).$$

For the antipodal shortcut geometry, Eq. (47) can be written as

$$\widetilde{F}_\rho(0, z) = \widetilde{\mathcal{F}}_\rho(0, z) + qz \widetilde{W}_\rho(L, z) \left[1 - \widetilde{\mathcal{F}}_L(0, z) \right] \widetilde{H}_{N, N/2}(z). \quad (58)$$

Substituting

$$\begin{aligned} \widetilde{\mathcal{F}}_\rho(0, z) &= \frac{T_{L-\rho}(y)}{T_L(y)}, & \widetilde{W}_\rho(L, z) &= \frac{U_{\rho-1}(y)}{zqT_L(y)}, \\ 1 - \widetilde{\mathcal{F}}_L(0, z) &= \frac{T_L(y) - 1}{T_L(y)}, & \widetilde{H}_{N, N/2}(z) &= \frac{T_L(y)}{D(y)} \end{aligned} \quad (59)$$

gives

$$\begin{aligned} \widetilde{F}_\rho(0, z) &= \frac{T_{L-\rho}(y)}{T_L(y)} + \frac{U_{\rho-1}(y) [T_L(y) - 1]}{T_L(y) D(y)} \\ &= \frac{T_{L-\rho}(y) D(y) + U_{\rho-1}(y) [T_L(y) - 1]}{T_L(y) D(y)} \\ &= \frac{aT_{L-\rho}(y) T_L(y) + T_{L-\rho}(y) U_{L-1}(y) + U_{\rho-1}(y) T_L(y) - U_{\rho-1}(y)}{T_L(y) D(y)}. \end{aligned} \quad (60)$$

The Chebyshev identity

$$T_L(y)U_{L-\rho-1}(y) = T_{L-\rho}(y)U_{L-1}(y) - U_{\rho-1}(y)$$

turns the numerator of Eq. (60) into

$$T_L(y) [aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)],$$

where $U_{-1} = 0$ when $\rho = L$. After cancellation of this common $T_L(y)$ factor, the expanded route gives

$$\tilde{F}_\rho(0, z) = \frac{aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)}{aT_L(y) + U_{L-1}(y)}. \quad (61)$$

Thus Eq. (45) is the corrected compact form of the expanded calculation rather than an additional assumption. In particular, following the full Eq. (41)-style hard route, represented above by Eq. (57), leads to the same cancellation: the apparent $T_L(y) = 0$, or α_ℓ , singularities cancel before the final pole set is read off.

This is not an argument that the numerator alone determines poles. The unreduced common denominator in Eq. (60) is $T_L(y)D(y)$, so the zeros of $T_L(y)$ are genuine candidate poles before the expression is collected. The point of the calculation is that those candidates are removable. To make the connection with the α -notation explicit, put $s = 1/z = 1 - q + qy$. If

$$\eta_\ell = \cos \frac{(2\ell - 1)\pi}{2L}, \quad \alpha_\ell = 1 - q + q\eta_\ell,$$

then

$$T_L(y(s)) = \frac{2^{L-1}}{q^L} \prod_{\ell=1}^L (s - \alpha_\ell).$$

Similarly,

$$U_{L-1}(y(s)) = \frac{2^{L-1}}{q^{L-1}} \prod_{r=1}^{L-1} (s - \gamma_r), \quad \gamma_r = 1 - q + q \cos \frac{r\pi}{L}.$$

Therefore the reduced denominator is proportional to the mixed polynomial

$$\Delta(s) = a \prod_{\ell=1}^L (s - \alpha_\ell) + q \prod_{r=1}^{L-1} (s - \gamma_r).$$

The α_ℓ -values enter this polynomial, but they are not individual denominator factors. Indeed, at $s = \alpha_m$,

$$\Delta(\alpha_m) = q \prod_{r=1}^{L-1} (\alpha_m - \gamma_r) \neq 0,$$

because the half-integer-angle α_m values do not coincide with the integer-angle γ_r values. Thus $s = \alpha_m$ is not a root of the final denominator.

Equivalently, let $y = \eta_\ell$ be a zero of $T_L(y)$, corresponding to an α_ℓ -mode location in z . The common-denominator numerator in Eq. (60) is

$$N_\rho(y) = aT_{L-\rho}(y)T_L(y) + T_{L-\rho}(y)U_{L-1}(y) + U_{\rho-1}(y)T_L(y) - U_{\rho-1}(y).$$

The Chebyshev identity gives

$$N_\rho(y) = T_L(y) [aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)],$$

so $N_\rho(\eta_\ell) = 0$. Therefore the apparent $1/T_L(y)$ pole has zero residue once the terms are collected, and the collected expression has no remaining $1/T_L(y)$ factor or double pole at $T_L(y) = 0$. Equivalently, the α_ℓ -modes that appear in the hard-route expansion are not poles of the final first-passage generating function.

APPENDIX C: TAIL AND NUMERICAL CHECKS

Let

$$N_\rho(y) = aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y).$$

The poles of the collected first-passage generating function are the roots of $D(y) = aT_L(y) + U_{L-1}(y)$. For simple roots $D(y_j) = 0$, $s_j = 1 - q + qy_j$,

$$F_\rho(0, t) = \sum_j B_{\rho j} s_j^{t-1}, \quad B_{\rho j} = \frac{qN_\rho(y_j)}{D'(y_j)}. \quad (62)$$

Thus the corrected long-time decay is controlled by the dominant root of D , not by a generic $t\alpha_1^t$ contribution.

C.1 Numerical audit protocol

The numerical audit is deterministic: it uses finite matrices and finite convolutions, not random simulation. The benchmark is the stochastic shortcut matrix itself. Starting from the lazy ring transition matrix, the shortcut matrix moves

$$\lambda = \beta(1 - q)$$

from the $u \rightarrow u$ self-loop to the directed $u \rightarrow v$ edge, so the u -row remains stochastic. The audit then uses the following independent checks.

1. **Resolvent benchmark.** It computes

$$\tilde{S}(z) = (I - z\tilde{T})^{-1}, \quad \tilde{F}_{n_0}(v, z) = \frac{\tilde{S}_{n_0, v}(z)}{\tilde{S}_{v, v}(z)},$$

and compares this direct finite-matrix ratio with both the corrected resolvent formula and the collected Chebyshev closed form.

2. **Killed-chain benchmark.** It deletes the absorbing target v and computes killed propagators directly from powers of the transient submatrix. This checks the corrected Eq. (14) bracket against the absorbing-chain definition, and also shows that the old reversed bracket has a visible finite-matrix error.
3. **Time-domain hard-route benchmark.** It computes the direct shortcut first-passage PMF from the shortcut matrix, and separately computes the corrected time-domain convolution route. For Eq. (57), it records the five displayed groups separately:

$$\mathcal{F}_{n_0}(v, \cdot), \quad qW * \delta * h_0\delta_0, \quad qW * \delta * H_+, \quad -qW * \mathcal{F}_u * h_0\delta_0, \quad -qW * \mathcal{F}_u * H_+.$$

The audit checks both that these five groups sum to the hard-route convolution and that the same sum matches the direct finite-matrix PMF at every checked time.

4. **Kernel-expansion benchmark.** It checks the displayed partial-fraction replacements for $A_t^{(+)}$, $B_t^{(0)}$, and $B_t^{(+)}$, including the diagonal limits used in the fully written Eq. (57)-style display. This verifies that the fully expanded display is algebraically the same as the kernel form of Eq. (57).
5. **Pole benchmark.** It computes the dominant eigenvalue s_1 of the finite transient shortcut matrix, then checks that $D(y) = aT_L(y) + U_{L-1}(y)$ is zero at $y = (s_1 - (1 - q))/q$, up to numerical precision. It also checks the sampled inequality $\gamma_1 < s_1 < \alpha_1$.
6. **Finite-time peak benchmark.** It directly iterates the finite shortcut chain for the $N = 100$, $q = 2/3$, $u = 6 \rightarrow v = 56$ setup and applies the same peak classifier to the resulting first-passage PMF.

The CSV outputs retain the direct benchmark columns, the formula columns, and the pointwise errors, so the table below is only the summary of the recorded audit rather than the audit itself.

The main checks are:

Check	Maximum error or result
Eq. (14), corrected bracket	1.18×10^{-14}
Eq. (14), old reversed bracket	2.00×10^{-1}
Orthogonality identities	6.34×10^{-13}
Hard-route convolution	6.94×10^{-18}
Eq. (57) five-group decomposition	5.20×10^{-18}
Kernel-to-fully-expanded identities	2.66×10^{-15}
Corrected resolvent formula	4.16×10^{-17}
Closed form	1.11×10^{-16}
Sampled pole inequality	$\gamma_1 < s_1 < \alpha_1$

The same run also reproduces the $N = 100$, $q = 2/3$, $u = 6 \rightarrow v = 56$ finite-time double-peak scan quoted in the discussion.

C.2 Sampled beta bounds for the $N = 100$ finite-time double peak

For the finite-time peak check, the comparison value is

$$\beta_c = \frac{2q}{(1-q)N} = 0.04 \quad (N = 100, q = 2/3).$$

This value is used as a reference line in the scan; it is not used as an analytic pole-crossing rule for the corrected tail. The corrected denominator check above shows instead that the dominant pole remains between γ_1 and α_1 on the sampled positive- β grid.

The scan uses the exact finite shortcut chain with target 56 and directed shortcut $6 \rightarrow 56$. The tested beta grid is

$$\{0, 0.002, 0.004, \dots, 0.080\} \cup \{0.001, 0.005\},$$

which also contains $\beta_c = 0.040$, $0.75\beta_c = 0.030$, and $1.25\beta_c = 0.050$. For each $n_0 = 1, \dots, 6$, the first-passage mass is iterated to $t_{\max} = 12000$ or survival mass $< 10^{-13}$. A curve is labelled as a *clear* double peak only when it has two local peaks above 10^{-12} , the second peak is at least 1% of the largest peak, the two peaks are separated by at least 10 time steps, the height ratio h_2/h_1 lies in $[0.1, 10]$, and the valley height is at most 0.8 of the smaller peak. This extra “clear” condition prevents adjacent early-time oscillations from being counted as a two-timescale double peak. The table reports the last sampled non-clear value before a clear interval, the first clear value, the last clear value, and the first sampled non-clear value after that interval.

n_0	no before	first clear	last clear	no after	clear at $\beta \geq \beta_c$
1	0.001	0.002	0.030	0.032	0
2	0.001	0.002	0.030	0.032	0
3	0.000	0.001	0.030	0.032	0
4	–	–	–	–	0
5	–	–	–	–	0
6	–	–	–	–	0

Thus, on this sampled grid, $n_0 = 1, 2$ have a clear double peak on $0.002 \leq \beta \leq 0.030$. The lower entry is bracketed by $0.001 < \beta \leq 0.002$, and the loss of the clear double peak is bracketed by $0.030 < \beta \leq 0.032$. For $n_0 = 3$, the sampled clear range is $0.001 \leq \beta \leq 0.030$, with lower entry bracketed by $0 < \beta \leq 0.001$ and the same upper-loss bracket $0.030 < \beta \leq 0.032$. For $n_0 = 4, 5, 6$, no clear double-peak interval is found anywhere in $0 \leq \beta \leq 0.080$. Local peak detectors can still find adjacent early-time bumps for these starts, but their first two peaks are only two time steps apart, so they fail the separation criterion and are not counted as clear two-timescale double peaks.